Note 7.2 Exponential Change and Separable Differential Equations

Separable Differential Equations

- **Differential Equations**
  It has the form
  $$\frac{dy}{dx} = f(x, y)$$
  where f is a function of both independent and dependent variables.
  A solution of this equation is a differentiable function $y = y(x)$ defined on the domain of x such that
  $$\frac{d}{dx}y(x) = f(x, y(x))$$
  on the interval. And the general solution is a solution $y(x)$ contains all solutions which always contains an arbitrary constant.
- **Separable Differential Equations**
  The differential equation that can be expressed as a product of a function of x and a function of y is separable. It has the form
  $$\frac{dy}{dx} = g(x)H(y)$$
  where g is a function of x and H is a function of y. Then we can rewrite this equation in the form
  $$\frac{dy}{dx} = \frac{g(x)}{h(y)}$$
  where $h(y) = \frac{1}{H(y)}$. That equals to
  $$h(y)dy = g(x)dx$$
  Now we can simply integrate both sides of this equation,
  $$\int h(y)\,dy = \int g(x)\,dx$$

Exercise Section 7.2

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